

Data Management & Personal Data Protection (DM & PDP) Strategy 2025-2027

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Approval

This strategy has been reviewed by the Data Governance Committee and approved by the Executive Leadership.

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1. EXECUTIVE SUMMARY

1.1 Overview

Data is the lifeblood of modern government administration. For Company A, a leading public sector entity in the Kingdom of Saudi Arabia, the effective management of data is not merely a compliance exercise but a strategic imperative to deliver world-class services to citizens, residents, and businesses.

This Data Management & Personal Data Protection (DM & PDP) Strategy (2025-2027) serves as the foundational blueprint for Company A's data journey over the next three years. It has been developed in strict alignment with the National Data Management Office (NDMO) regulations, the Personal Data Protection Law (PDPL), and the broader goals of Saudi Vision 2030.

The strategy articulates how Company A will transition from a "Compliance-Focused" organization to a "Value-Driven, AI-Ready" entity. It encompasses all 15 domains of the NDMO National Data Management Framework, detailing specific initiatives to govern, protect, catalog, and leverage data assets.

With an approved budget of SAR 3.5 Million over three years and a dedicated team of 12 data professionals, Company A is poised to set a benchmark for data excellence in the government sector.

1.2 Key Achievements (2023-2024)

Before outlining the future, it is critical to acknowledge the progress made during the previous strategic cycle (2023-2024), labeled the "Establishment Phase."

- Governance Establishment: Successfully established the Data Management Office (DMO) and the Data Governance Committee (DGC), which has met 24 times since inception.
- Compliance Score: Raised the NDMO compliance score from an initial 45% (2022) to 92% (Q4 2024), placing Company A in the top tier of government entities.
- Data Cataloging: Implemented the Enterprise Data Catalog (Informatica EDC), successfully scanning and registering 100% of Critical Systems (58 Data Sources) and cataloging over 4,500 data assets.
- Data Quality: Deployed automated data quality rules for the top 50 Critical Data Elements (CDEs), achieving an average quality score of 92%.
- Privacy Compliance: Appointed a Chief Privacy Officer (CPO) and implemented a Consent Management Platform, processing 150,000+ consents in compliance with the PDPL.
- Open Data: Published 18 high-value datasets on the National Open Data Portal, driving transparency and research utility.

1.3 Strategic Outlook (2025-2027)

The 2025-2027 cycle, labeled the "Value Creation Phase," focuses on three main themes:

1. Optimization (2025): Perfecting the basics. We aim to reach 98% compliance with NDMO Priority 1 specifications, automate 100% of data quality checks for critical data, and fully integrate our metadata repository with business processes.
2. Intelligence (2026): democratizing data. We will rollout self-service Business Intelligence (BI) capabilities to 500+ employees, reducing the "time-to-insight" from days to minutes.
3. Innovation (2027): AI readiness and monetization. We will build the foundations for Generative AI adoption, ensuring our data is "AI-ready" (clean, labeled, and governed), and explore secure data monetization/sharing models with the private sector.

Summary of Key Targets by 2027:

- NDMO Compliance: Sustained >96%.
- Data Quality Score: >98% for all Critical Data Elements.

- Data Literacy: 100% of staff certified in basic data awareness.
- Data Maturity: Reach NDMO Maturity Level 5 (Optimized).

2. STRATEGIC CONTEXT

2.1 Organization Overview

Company A operates within the [Healthcare/Education/Finance] sector, providing essential services to a constituent base of over 2 million individuals.

- Employees: 850 Full-Time Staff.
- Digital Footprint: 58 Integrated Systems, including Oracle ERP, SAP HR, and Custom E-Services Portals.
- Data Volume: Managing approximately 450 TB of structured and unstructured data, growing at 20% annually.
- Key Stakeholders:
 - Citizens: Expecting personalized, seamless digital services.
 - Regulators (NDMO/NCA/SDAIA): Expecting strict adherence to sovereignty and security laws.
 - Partners (MOH/MOF/NIC): Expecting reliable, real-time data exchange (G2G).

2.2 Vision 2030 Alignment

This strategy is not an isolated document but a direct contributor to the Kingdom’s Vision 2030 goals:

Vision 2030 Theme	Company A Data Strategic Goal
A Vivid Society	Protecting Privacy: Ensuring the personal data of
A Thriving Economy	Open Data: Publishing non-sensitive data to fuel t
An Ambitious Nation	Government Effectiveness: Using data analytics to

2.3 NDMO Mandate & Compliance Requirements

The National Data Management Office (NDMO) has issued 15 Data Management Domains and 191 Controls. As of Q4 2024, Company A’s compliance status is as follows:

Priority Level	Description	Target 2024	Actual Q4 2024	Target 2025
Priority 1 (P1)	Foundational capabilities (Governance, Security, P	100%	100%	100%
Priority 2 (P2)	Advanced capabilities (Catalog, Quality, Architect	95%	91%	95%
Priority 3 (P3)	Mature capabilities (Monetization, Document Mgmt).	90%	83%	90%

Critical Gaps to Address in 2025:

1. Unstructured Data Governance (P3): Digitizing and cataloging physical archives remains at 60% completion.
2. Master Data Management (P2): While "Customer" and "Product" domains are mastered, "Vendor" and "Location" data remain siloed.
3. Data Valuation (P3): No formal financial value has been assigned to data assets.

2.4 SWOT Analysis (Data Capabilities)

Strengths (Internal)	Weaknesses (Internal)
• Strong Leadership Support: CEO and Board are act	• Legacy Systems: 3 core systems (Finance) are 10+
Opportunities (External)	Threats (External)
• AI & GenAI: Leveraging clean data for predictive	• Cyber Threats: Increasing sophistication of rans

3. STRATEGIC DIRECTION

3.1 Vision Statement

> "To be the trusted model of data excellence in the Saudi public sector, empowering Company A to deliver proactive, personalized, and efficient services through secure, high-quality, and accessible data."

3.2 Mission Statement

> "To implement and sustain a world-class data management office that ensures the accuracy, availability, protection, and ethical use of Company A's data assets, fostering a culture of evidence-based decision making in alignment with National Sovereignty."

3.3 Guiding Principles

1. Data is an Asset: We manage data with the same rigor as financial and physical assets.
2. Privacy by Design: We consider data protection at the start of every project, not at the end.
3. Quality at Source: We fix data issues where they are created, not in the reports.
4. Open by Default: Unless classified as Restricted, data should be shared to maximize value.
5. Standardized: We speak one data language across all departments (One Truth).

3.4 Strategic Pillars

The strategy is built upon four robust pillars:

- 1. GOVERNANCE: Ensuring accountability, clear roles, and compliance.
- 2. OPERATIONS: Efficiently managing the lifecycle, quality, and flow of data.
- 3. PROTECTION: Safeguarding data privacy and security against threats.
- 4. VALUE: Extracting insight and economic value through analytics and sharing.

4. STRATEGIC OBJECTIVES & INITIATIVES (BY DOMAIN)

4.1 Domain 1: Governance & Leadership

Objective: Maintain a robust, sustainable governance framework that survives personnel changes.

Initiative ID	Initiative Name	Description	KPI	Owner
DG-2025-01	Policy Automation	Integrate Policy enforcement into CI/CD pipelines	100% Policy Deployments Checked	CDO
DG-2025-02	Stewardship 2.0	Launch a gamified "Data Steward" certification	200 Certified Stewards	Governance Lead
DG-2025-03	Architecture Board	Establish a Data Architecture Review Board (DARB)	100% Board Approved	Lead Architect

Implementation Detail: The Policy Automation initiative involves configuring the 'Archer' GRC tool to speak directly to 'Jenkins' pipelines. Requires 3 months of API development.

4.2 Domain 2: Data Catalog & Metadata

Objective: Make data discoverable and understandable for all 800+ employees.

Initiative ID	Initiative Name	Description	KPI	Owner
DC-2025-01	Lineage Complete	Map column-level lineage for all PII data	100% Regulatory Lineage	Catalog Admin
DC-2025-02	Business Glossary	Expand glossary to include 1,000+ terms	1000+ Terms ver	Governance Lead
DC-2025-03	Data Marketplace	Launch an internal "Data Shop" for requests	500 Requests/Month	CDO

4.3 Domain 3: Data Quality & Trust

Objective: Ensure "Zero Defect" data for critical decision-making.

Initiative ID	Initiative Name	Description	KPI	Owner
DQ-2025-01	Address Cleanup	Complete remediation of National Address	100% Valid Addresses	DQ Lead

DQ-2025-02	Preventive DQ	Implement "Firewall for Data" - AP	Reduce data entry errors	IT Director
DQ-2025-03	AI-Driven DQ	Pilot AI tool to auto-suggest DQ rules	Pilot Data Self-heal	DQ Lead

- Target Metrics:
- Completeness: 99.5%
 - Accuracy: 98%
 - Uniqueness: 100% for Customer Profiles

4.4 Domain 5 & 15: Security & Protection

Objective: Zero breaches and full alignment with NCA ECC.

Initiative ID	Initiative Name	Description	KPI	Owner
SEC-2025-01	Dynamic Masking	Implement dynamic data masking (DDM) for PII	100% PII Data Masked	CISO
SEC-2025-02	DLP Expansion	Extend Data Loss Prevention rules to all cloud storage	100% Coverage	CISO

4.5 Domain 6 & 14: Privacy & Ethics (PDPL)

Objective: Be the gold standard for Privacy compliance in government.

Initiative ID	Initiative Name	Description	KPI	Owner
PDP-2025-01	Privacy Automation	Deploy "OneTrust" or similar to automate PDPA Subj	95% Automated	CPO
PDP-2025-02	Consent Audit	Re-verify all legacy consents against new regulations	100% Valid Consent	CPO

4.6 Domain 8: Analytics & Value Creation

Objective: Shift from "Reporting on the Past" to "Predicting the Future".

Initiative ID	Initiative Name	Description	KPI	Owner
AN-2025-01	Self-Service BI	Train 50 "Power Users" in Tableau/Power BI	50 Certified Users	Analytics Lead
AN-2025-02	Predictive Pilot	Build 1 predictive model (e.g., Predicting Service Requestionized	Model Accuracy	Data Scientist

5. IMPLEMENTATION ROADMAP (2025-2027)

5.1 Phase 1: Optimization (2025)

Theme: "Solidifying the Foundation"

- Q1: Launch "Preventive DQ" API controls; Complete 100% User Access Review.
- Q2: Rollout "Data Marketplace" internally; Conduct annual Privacy Impact Assessment (PIA).
- Q3: Achieve 95% NDMO Compliance in annual audit; Migrate Vendor Master Data to MDM.
- Q4: Complete "Dynamic Masking" project; Publish Annual Data Report.

5.2 Phase 2: Intelligence (2026)

Theme: "Speed to Insight"

- Q1: Deploy Self-Service BI Platform enterprise-wide.
- Q2: Integrate with National Data Bank (NDB) for 5 key datasets.
- Q3: Implement AI-based Anomaly Detection for Security.
- Q4: Pilot "Predictive Demand Planning" model for Operations team.

5.3 Phase 3: Innovation (2027)

Theme: "AI & Value"

- Q1: Establish "Generative AI Center of Excellence".
- Q2: Launch "Open Data Hackathon" to encourage public use of our data.
- Q3: Monetize 3 API data services (if regulation permits).
- Q4: Full automation of Governance (Governance-as-Code).

6. GOVERNANCE & OPERATING MODEL

6.1 Organization Structure

Evidence of Maturity: Company A has established a dedicated Data Management Office (DMO) reporting directly to the CDO, who peers with the CIO.

- Total Headcount: 12 FTEs
- Functions:
 - Governance & Privacy: 3 FTEs (Policy, Compliance, PDPL)
 - Quality & Architecture: 4 FTEs (DQ Rules, Models, Catalog)
 - Engineering & Ops: 3 FTEs (ETL, Database, Backup)
 - Analytics: 2 FTEs (Dashboards, AI)

6.2 Roles & Responsibilities

Chief Data Officer (Ahmed Al-Rashid)

- Strategic: Defines vision and strategy.
- Fiduciary: Managing the SAR 3.5M budget.
- Representative: Represents Company A at NDMO and SDAIA.

Chief Privacy Officer (Noura Al-Otaibi)

- Legal: Interpretation of PDPL.
- Operational: Handling breach notifications and subject requests.
- Auditing: Ensuring Privacy by Design in all IT projects.

Business Data Executive (e.g., HR Director)

- Accountability: Owner of all HR data assets.
- Approval: Signs off on access requests to HR data.
- Resource: Dedicates 20% of HR Data Steward's time to governance.

Data Steward (Operational Role)

- Definition: Writes business definitions in the Glossary.
- Quality: Fixes DQ errors assigned to them.
- Change: Approves changes to data structures.

6.3 Decision Rights Matrix

Decision Area	Decisions	Key Decision Maker	Consulted
Budget	Approval of Data Tool purchases >50k	DGC	CIO, CFO
Policy	Approval of new Data Policies	DGC	Legal, HR
Access	Granting access to Secret data	Data Owner + CISO	Line Manager
Standard	Adoption of a new naming convention	CDO	Architecture Lead
Issue	Resolution of cross-dept DQ conflict	DGC	Business Execs

7. BUDGET & RESOURCE PLANNING

7.1 Financial Budget (2025)

Total Requested: SAR 1,200,000

Category	Item Description	Cost (SAR)	Justification
Software Licenses	Informatica EDC/IDQ Renewal	450,000	Critical for P2 compliance.
Consulting	NDMO Annual Audit (External)	200,000	Required independent verification.
Training	CDMP Certification for 5 staff	100,000	Skill upskilling.
Staffing	1 Senior Data Scientist (Hiring)	300,000	For Predictive Pilot.
Awareness	Data Privacy Day Events	50,000	Culture building.
Contingency	Unforeseen needs	100,000	Risk buffer.

7.2 Human Resources Plan

- Recruitment: Hiring 1 Data Scientist in Q1 2025.
- Retention: Introducing "Critical Skill Bonus" for Data Engineers to reduce turnover.
- Succession: DMO Manage designated as successor to CDO.

7.3 Technology Stack

- Governance: Archer GRC.
- Catalog: Informatica EDC.
- Quality: Informatica IDQ.
- Visualization: Microsoft PowerBI Premium.
- Database: Oracle Exadata (On-prem) + Azure Synapse (Analytics).

8. PERFORMANCE MANAGEMENT (KPIs)

8.1 Strategic KPIs (Board Level)

KPI	Metric Definition	2024 Baseline	2025 Target
S-01	NDMO Compliance Score	92%	96%
S-02	Data Quality Index (CDEs)	92%	95%
S-03	Data Literacy Rate	85%	95%
S-04	Value Realized (Savings)	SAR 500k	SAR 1.2M

8.2 Operational Metrics (DMO Level)

Metric	Frequency	Owner	Target
Glossary Terms Defined	Monthly	Gov Lead	+50 / Month
Catalog Active Users	Monthly	Catalog Admin	> 300 Users
DQ Issues Open >30 Days	Weekly	DQ Lead	0
Access Requests SLA	Monthly	Security	< 24 Hours
Privacy Requests SLA	Monthly	CPO	< 5 Days

9. RISK MANAGEMENT

9.1 Strategic Data Risks

Risk ID	Risk Description	Prob	Impact	Risk Level
R-STR-01	Talent Loss: Key data staff leaving for private se	High	High	Critical
R-STR-02	Cyber Breach: Ransomware attack on Patient DB.	Critical	Critical	High
R-STR-03	Regulatory Change: New unforeseen laws requiring c	Med	Med	High
R-STR-04	Low Adoption: Business ignores Med catalog and us	Med	Med	Med

9.2 Mitigation Strategies

- For Talent Loss: Implemented retention bonuses and clear career paths.
- For Cyber Breach: Implemented Immutable Backups (Object Lock) and Air-gapped recovery options.
- For Low Adoption: Launched "Data Champions" network to evangelize tools in business units.

10. APPENDIX

10.1 Glossary of Terms (Sample of 100+)

- Availability: Ensuring timely and reliable access to and use of information.
- Critical Data Element (CDE): Data that is vital to the operation of the organization.
- Data Catalog: An inventory of data assets in the organization.
- Data Custodian: IT resource responsible for safe custody, transport, and storage of data.
- Data Owner: Senior business executive accountable for specific data.

- Data Steward: Business subject matter expert responsible for data meaning and quality.
- Governance: The exercise of authority and control (planning, monitoring, and enforcement).
- Metadata: Data about data (e.g., who owns it, where it comes from).
- NDMO: National Data Management Office.
- PDPL: Personal Data Protection Law.
- Personal Identifiable Information (PII): Data that can identify an individual.
- Processing: Any operation performed on personal data.
- Processor: Entity processing data on behalf of controller.
- Profiling: Automated processing to evaluate personal aspects.
- Pseudonymization: Processing to prevent direct identification.
- Quality Assurance: Systematic monitoring to ensure data standards.
- Record: Documented information in any form.
- Reference Data: Data used to classify other data.
- Remediation: Actions to address non-compliance.
- Repository: Storage location for data assets.
- Retention: Duration for which data is kept.
- Risk Assessment: Process to identify and evaluate risks.
- Role-Based Access (RBAC): Access granted based on user role.
- Schema: Structure defining data organization.
- Security Controls: Measures to protect data.
- Segregation of Duties: Separation of conflicting responsibilities.
- Sensitivity Level: Degree of confidentiality required.
- Service Level Agreement (SLA): Commitment to service standards.
- Single Source of Truth (SSOT): Authoritative data source.
- Stakeholder: Individual with interest in data management.
- Standard: Documented specification for consistency.
- Stewardship: Accountability for data quality and meaning.
- Storage: Physical or logical location of data.
- Strategy: Long-term plan for data management.
- Structured Data: Data organized in defined format.
- Subject Matter Expert (SME): Person with specialized knowledge.
- System of Record (SOR): Authoritative source for data element.
- Technical Debt: Accumulated suboptimal technical decisions.
- Third Party: External entity accessing data.
- Threat: Potential cause of unwanted incident.
- Timeliness: Degree to which data is current.
- Traceability: Ability to trace data lineage.
- Training: Education on data management practices.
- Transaction: Business event recorded in data.
- Transformation: Process of converting data format.
- Transparency: Clarity in data processing activities.
- Trustworthiness: Reliability and accuracy of data.
- Unstructured Data: Data without predefined structure.
- Validation: Verification of data accuracy.
- Value: Benefit derived from data assets.

- Vendor: External supplier of services or products.
- Version Control: Management of document versions.
- Vulnerability: Weakness that can be exploited.
- Workflow: Series of connected steps in process.

10.2 Reference Documents

1. NDMO National Data Management Framework V3.0
2. Personal Data Protection Law (Royal Decree M/19)
3. NCA Essential Cybersecurity Controls (ECC-1:2018)
4. Company A Strategic Plan 2023-2027

END OF DOCUMENT

Company A - Internal Use Only